

Harsh Shah

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Summary

Software Engineer with 6+ years of experience delivering enterprise-scale cloud-native applications across financial services, Generative AI, e-commerce and IoT domains. Experienced building full-stack applications, scalable backend services and distributed systems using React, Angular, TypeScript, Node.js, Java, C#, Python, AWS and Azure. Recent experience building GenAI, RAG pipelines and agentic workflows, with proven ability to own features end-to-end from frontend architecture and API development to cloud-deployed microservices.

Technical Skills

Languages: Java, TypeScript, JavaScript, C#, Python, Kotlin, C++, Go

Frontend/Mobile: React, Angular, Next.js, Vue.js, HTML5, CSS, SCSS, Tailwind CSS, Magento, React Native, Android

Backend/API: Node.js, .NET, Spring Boot, GraphQL, REST APIs, Express, FastAPI

Cloud/DevOps: AWS (Lambda, API Gateway, S3, Bedrock), Azure (Functions, Event Hub, AD B2C), Docker, Kubernetes, CI/CD, GitHub Actions

AI/ML: RAG, AI Agents, LLM Applications, Vector Databases, LangChain, LangSmith, OpenAI API, Claude API

Databases: PostgreSQL, SQL Server, MongoDB, Redis, DynamoDB, Pinecone, OpenSearch, pgvector

Testing/Practices: Git, GitLab, Jest, React Testing Library, Cypress, Agile, Scrum, Jira

Professional Experience

Valtech | *Detroit, MI, USA*

June 2021 – Present

Software Engineer

- Built a production **GenAI Document Q&A** application using **React**, **TypeScript**, **Claude API** and Retrieval Augmented Generation (**RAG**) architecture to deliver accurate, citation-backed responses from enterprise documents
- Implemented **Human-in-the-Loop** (HITL) review workflows with citation highlighting, response validation and override capabilities, **agent reasoning visualization** and live **LangSmith** tracing reducing document review time by **85%**
- Designed and developed a real-time **Angular** IoT dashboard monitoring **1,000+** LoRa gateways, visualizing telemetry with **Highcharts Stock** and implementing multi-tenant role-based access using **Azure AD B2C**
- Built the **C#/.NET Azure Functions** backend processing telemetry ingestion into **Event Hub** and **Service Bus**, with automated fault detection and event-driven workflows triggering push, email and SMS alerts through **Azure Notification Hubs** to improve operational visibility and incident response
- Developed an enterprise reporting platform using **React** and **Java**, building RESTful services and a reporting engine with **AG Grid Enterprise**, custom PDF generation, filtering and XLSX export across **20+** county tax and loan management workflows
- Modernized a **Sitecore** based web platform using **Angular**, **Node.js** and **EJS** templates improving data table filtering and cutting page load times by **40%**
- Delivered a Magento B2B e-commerce platform for **2,000+** HVAC contractors, adding custom modules, **Azure AI Search** autocomplete, and a bulk CSV Quick Order flow with Paymetric checkout that reduced distributor ordering time by **90%**
- Developed and maintained a production **React Native/Expo** app with **10,000+** active users, built geofencing automation and multi-thermostat control backed by **GraphQL** and **Apollo Client** on **AWS** serverless infrastructure

Ascent FutureTech | *Mumbai, MH, India*

July 2017 – July 2018

Associate Software Engineer

- Built a **Python/Flask REST API** backend for service listing, bidding and contracts for a freelancer marketplace; integrated the Braintree payment gateway with **webhook handling**, contributing to a **16%** increase in successful transactions
- Developed an inventory management dashboard (**Angular**, **Spring Boot**) for order creation, status tracking and delivery workflows over a relational backend

Education

Northeastern University | *Boston, MA, USA*

September 2018 – December 2020

Master of Science in Information Systems

Mumbai University | *Mumbai, MH, India*

July 2013 – June 2017

Bachelor of Engineering in Information Technology

Independent Projects

RAG Document Intelligence Agent (Python, FastAPI, LangChain, pgvector, Claude API, RAGAS)

- Built a **RAG** pipeline ingesting PDFs via **PyMuPDF** chunking with recursive splitting and OpenAI embeddings into **pgvector** (Supabase), with hybrid BM25 + vector retrieval fused via RRF improving context recall from **0.64 to 0.78** over pure vector search
- Implemented **agentic tool calling** via the **Claude API** with LangSmith tracing and XML-structured citation prompts, reaching a **faithfulness score of 0.84** on a 70-item eval dataset
- Built **RAGAS** regression suite measuring answer relevancy, context precision and used eval sets to tune chunking strategy